Adaptation Experiment 2: Retrieval Baselines

1 Signals and leakage-free retrieval

For query date s, the exact window convention is

$$X_s = (s - L, s], \qquad Y_s = (s, s + H].$$

Let $V = f(X_s)$ be the vanilla backbone forecast and let C be the same backbone conditioned on retrieved examples. For neighbor j, let r_j be its last look-back date, Y_j its observed future, and $N_j = f(X_{r_j})$ its vanilla forecast. All neighbor values are transferred to the query's level and scale before use. Distance weights are

$$w_j = \frac{\exp(-\widetilde{d}_j)}{\sum_{\ell=1}^{k} \exp(-\widetilde{d}_\ell)}, \qquad \widehat{Y}_h = \sum_{j=1}^{k} w_j Y_{j,h}.$$

In period-aligned retrieval, the last dates of the look-backs satisfy $s - r_j \equiv 0 \pmod P$. In fixed mode every (X_{r_j}, Y_{r_j}) lies wholly inside T_0. In online mode a neighbor is eligible only when $r_j + H \le s$, so its complete future is observed by query time. A retrieved future may overlap the query look-back, which is already observed, but it can never overlap Y_s. These conditions prevent target leakage.

Experimental goal. Compare all useful direct combinations of the available information V, C, Y_j, N_j before introducing a neural adapter.

2 Anchored models

Every fitted forecast is written directly as $V +$ linear signals. There is no intercept and no design based only on differences such as $Y_j - N_j$. The word "residual" below is only the agreed name for the model containing both Y_j and N_j.

A *shared* model has one coefficient vector used at every horizon. A *horizon* model fits H independent regressions. Each of those H regressions has the same number of coefficients as the shared regression; the total coefficient count is therefore multiplied by H.

Artifact name	Forecast at horizon h	Coefficients
`vanilla`	V_h	0
`context_forecast`	C_h	0
`aggr_y`	$\widehat{Y}_h$	0
`y_mean`	$k^{-1} \sum_j Y_{j,h}$	0
`aggr_y_mix_shared`	$(1-\lambda)V_h + \lambda \widehat{Y}_h$	1
`aggr_y_mix_horizon`	$(1-\lambda_h)V_h + \lambda_h \widehat{Y}_h$	1 per h
`context_ridge_shared`	$V_h + aV_h + bC_h$	2
`context_ridge_horizon`	$V_h + a_h V_h + b_h C_h$	2 per h
`aggr_y_ridge_shared`	$V_h + aV_h + b\widehat{Y}_h$	2
`aggr_y_ridge_horizon`	$V_h + a_h V_h + b_h \widehat{Y}_h$	2 per h
`y_ridge_shared`	$V_h + aV_h + \sum_j b_j Y_{j,h}$	$1 + k$
`y_ridge_horizon`	$V_h + a_h V_h + \sum_j b_{j,h} Y_{j,h}$	$1 + k$ per h
`cov_y_ridge_shared`	$V_h + aV_h + bC_h + \sum_j c_j Y_{j,h}$	$2 + k$
`cov_y_ridge_horizon`	$V_h + a_h V_h + b_h C_h + \sum_j c_{j,h} Y_{j,h}$	$2 + k$ per h
`cov_horizon_ridge_shared`	$V_h + aV_h + bC_h + c\widehat{Y}_h$	3
`cov_horizon_ridge_horizon`	$V_h + a_h V_h + b_h C_h + c_h \widehat{Y}_h$	3 per h
`residual_ridge_shared`	$V_h + aV_h + \sum_j b_j Y_{j,h} + \sum_j c_j N_{j,h}$	$1 + 2k$

Artifact name	Forecast at horizon h	Coefficients
`residual_ridge_horizon`	$V_h + a_h V_h + \sum_j b_{j,h} Y_{j,h} + \sum_j c_{j,h} N_{j,h}$	$1 + 2k$ per h
`full_ridge_shared`	$V_h + a V_h + b C_h + \sum_j c_j Y_{j,h} + \sum_j d_j N_{j,h}$	$2 + 2k$
`full_ridge_horizon`	$V_h + a_h V_h + b_h C_h + \sum_j c_{j,h} Y_{j,h} + \sum_j d_{j,h} N_{j,h}$	$2 + 2k$ per h

The two lambda models are the only retained convex mixtures. Their least-squares estimate is projected onto $[0, 1]$ with $\lambda \leftarrow \min(1, \max(0, \lambda_{\mathrm{LS}}))$; the horizon variant does this independently for each λ_h. Ridge coefficients are unconstrained.

3 Model-local split and validation protocol

Extraction fixes only

$$T_0 \ (30\%), \qquad T_1 + T_2 \ (50\%), \qquad T_3 \ (20\%),$$

and writes `adapt` and `eval` payloads. For these baselines, `adapt` is split chronologically by whole query dates, with its final 20% assigned to T_2 by default.

1. Accumulate exact float64 sufficient statistics on T_1.
2. For every ridge family, fit every $\alpha \in \{0, 10^{-6}, 10^{-5}, 10^{-4}, 10^{-3}, 10^{-2}, 10^{-1}, 1, 10\}$ and select the value with lowest T_2 nMSE.
3. Refit that family on all of T_1 +T_2 using the selected α.
4. Fit the lambda models directly on T_1 +T_2 because they have no regularization hyperparameter to validate.
5. Score every selected model once on the complete untouched T_3.

Features are RMS-standardized without centering, so a zero coefficient still means no added signal. Chunked sufficient statistics change memory usage, not the fitted objective. Optional fit-only sample caps never subsample final T_3 scoring. Optimistic `eval_fit` variants remain available behind an explicit flag for appendix diagnostics and are disabled by default.

Code: `src/experiments/extraction.py`; `src/adaptors/baselines/evaluate.py`.
Launchers: `extraction.slurm`; `baselines.slurm`; `horizon_baselines_ablation.slurm`.

4 Screening and ablation sequence

The candidate screen uses Chronos-2 on

$$\mathcal{D} = \{\text{Electricity, Traffic, Solar, Exchange}\}, \qquad \mathcal{S} = \{168{:}24, 336{:}48, 504{:}168\},$$

with $k \in \{1, 3\}$ and both raw and instance-normalized Euclidean retrieval. Fourier-amplitude retrieval is also implemented as an optional ablation space; it standardizes each look-back before taking the FFT magnitude and does not enter the primary screen. The selected-winner full/ultra benchmark retains these settings without reopening candidate selection. The primary screen contains the direct forecasts and every shared fitted baseline in the table above. It excludes every `*_horizon` baseline; those models are not candidates for the main comparison. Here a setting is one dataset–(L, H) pair, giving $|\mathcal{D}||\mathcal{S}| = 12$ settings. A candidate model is the complete pipeline (formula, retrieval normalization, k). Each such pipeline is ranked by its unweighted mean of the 12 setting-level percentage improvements over V; there is no averaging over k or normalization. Complete winner names are copied manually from the ranking for three separate studies:

$$\begin{array}{lll} k \text{ ablation:} & k \in \{1, 3, 5, 10, 15, 20\}, \quad (\mathcal{D}, \mathcal{S}), & \text{retaining formula/normalization,} \\ H \text{ ablation:} & L = 504, \quad H \in \{24, 168, 504\}, & \text{retaining formula/normalization}/k, \\ L \text{ ablation:} & H = 24, \quad L \in \{24, 168, 504\}, & \text{retaining formula/normalization}/k. \end{array}$$

The companion positive-window table averages, with the same configuration weighting, the saved percentage of complete windows whose horizon-averaged MSE is strictly lower than vanilla; no per-window metric vector is stored. Per-horizon parameterization is a separate ablation. The root front `horizon_baselines_ablation.slurm` accepts complete shared winners in `SHARED_BASELINE_WINNERS_CSV`; for every selected pipeline it evaluates both the shared model and its corresponding `*_horizon` model under the same retrieval normalization and k. The current defaults retain the best shared convex mixture, aggregated-target ridge, and individual-Y ridge from the completed screen. Its results are written under the distinct `horizon_baselines_ablation` profile and are reported only as an ablation.

The four original ETT panels and Weather are excluded from $\mathcal{D}$ because their channels mix unlike physical quantities; nearest neighbors across those channels may be semantically irrelevant even after scale normalization. The separate root front `mixed_quantity_ablation.slurm` transfers only selected complete screen winners to

$$\mathcal{D}_{\text{mixed}} = \{\text{ETTh1, ETTh2, ETTm1, ETTm2, Weather}\}$$

and writes a distinct `mixed_quantity_ablation` result profile. These results are an ablation and never contribute to primary winner selection. All variates are retained for these original ETT panels and Weather, as well as for Exchange in the primary set; constant-window counts are reported but are not converted into dataset-level exclusions.

Every ablation consumes selected complete screen winners. The k ablation is the only in-project exception to the ordinary $k \in \{1, 3\}$ policy; its wider grid is evaluated only for those selected pipelines. Cross-RAG separately retains the released method's prescribed $k = 15$ comparison.

The final `benchmark.slurm` comparison transfers only selected complete screen winners to $\mathcal{D}$ plus `ETT_T_1H`, `ETT_L_1H`, `ETT_T_15T`, and `ETT_L_15T`, while retaining the same $\mathcal{S}$. The selected winner name fixes its formula, normalization, distance, k, and retrieval mode. Full uses Chronos-2; ultra adds backbones without changing datasets, settings, or candidate pipelines. These merged ETT panels retain only one physical quantity; 15-minute extraction uses a 96-step daily retrieval period.

The Cross-RAG comparison is not crossed with $\mathcal{S}$: it separately uses Chronos-Bolt, $L = 512$, $H = 64$, $k = 15$, and min–max/cosine X-space retrieval. Only the winning in-project candidate is fitted in that profile.

5 Interpretation

`aggr_y` tests analog forecasting directly; `context` tests whether the backbone can consume retrieved context. The individual-Y models test whether learned neighbor-specific weights beat the fixed distance aggregation. `residual` tests whether neighbor forecasts add information about neighbor reliability, and `full` is the complete V, C, Y_j, N_j model. All claims must be based on the same complete T_3 queries and compared with V.